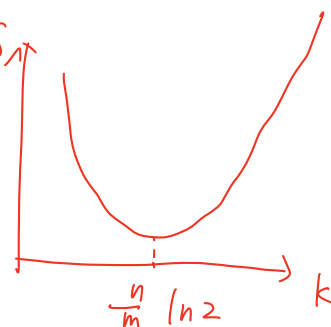


1 Sampling and Bloom Filter

Answer the following questions:

1. What are the key steps of Reservoir sampling with size $s > 1$?
2. What are the key steps of constructing a Bloom Filter?
3. Let k be the number of hash functions in a Bloom Filter. Suppose the number of bits in the bit array B and the number of elements in the key set S are fixed. What will happen to the false positive rate when k increases from 0 to ∞ ? Explain the reason in 2–3 sentences.

1. $(S > 1)$ ① put S items into Reservoir. ② each item m have



2. k independent hash function.

① set b all 0s

② hash each member $s \in S$

to n buckets use k hash function. , Set $B[h_i(s)] = 1$

① Initialize a bit array B of length n with all entries set to 0.

$0 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0$

② Choose k independent hash functions: $h_1, h_2, \dots, h_k$

③ Insert element x , compute $h_i(x)$ for each hash and then we set $B[h_i(x)] = 1$

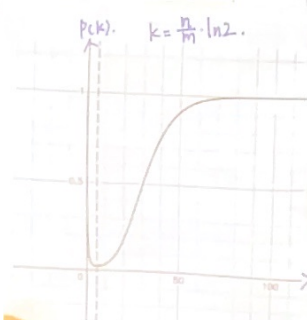
$0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 1$

Query element y .

If any $B[h_i(y)] = 0$, element y is definitely NOT in the set
 If any $B[h_i(y)] = 1$, y is possibly in the set

and the number of elements in the key set S are fixed. What will happen to the false positive rate when k increases from 1 to ∞ ? Explain the reason in 2–3 sentences.

1. ① put the first S items of stream into the Reservoir
- ② For each subsequent item m ($m > S$)
 - select it with probability $\frac{S}{m}$
 - If selected, replaced a random element currently in the Reservoir.



2. ① Initialize a bit array B of length n with all entries set to 0.

$0 \ 0 \ 1 \ 1 \ 1 \ 1 \ 0$

② Choose k independent hash functions: $h_1, h_2, \dots, h_k$

③ Insert element x , compute $h_i(x)$ for each hash and then we set $B[h_i(x)] = 1$

$0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 1$

④ Query element y .

If any $B[h_i(y)] = 0$, element y is definitely NOT in the set
 If any $B[h_i(y)] = 1$, y is possibly in the set

$p(k) = (1 - e^{-\frac{km}{n}})^k$

$k \in (0, \frac{n}{m} \ln 2)$, $p(k) \downarrow$

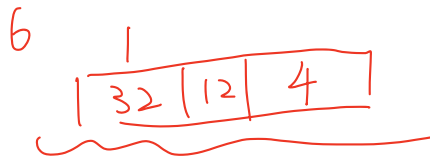
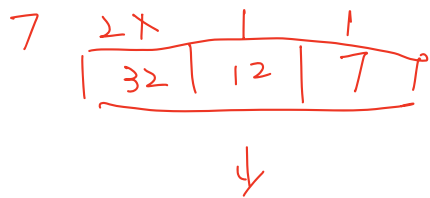
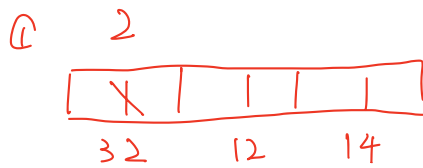
$k \in (\frac{n}{m} \ln 2, +\infty)$, $p(k) \uparrow$

m : elements inserted
 n : length of bit array

2 Misra-Gries Algorithm

{32, 12, 14, 32, 7, 12, 32, 7, 6, 12, 4}

- Given the data stream above, perform the Misra-Gries algorithm with $k = 3$ and report the summary.
- What expected number of times has the Misra-Gries summary triggered the decrement steps after processing the given stream? Please derive your answer as a function of m and k .



② size k .
element m .

m' is the number of item left in algorithm.

n = decrement step.

$$m = m' + (k+1)n.$$

$$n = \frac{m - m'}{k+1}$$

- Given the data stream above, perform the Misra-Gries algorithm with $k = 3$ and report the summary.
- What expected number of times has the Misra-Gries summary triggered the decrement steps after processing the given stream? Please derive your answer as a function of m and k .

1. $k=3$

$a_1 = 32$ arrives

Item	32
Count	1

$a_2 = 12$

1	32	12
C	1	1

$a_3 = 14$

1	32	12	14
C	1	1	1

$a_4 = 32$

1	32	12	14
C	2	1	1

$a_5 = 7$

1	32	12	14
C	2	1	1

"7" discarded

↓ ↓ ↓

0 0 0

$a_6 = 12$

1	32	12
C	1	1

$a_7 = 32$

2	32	12
C	2	1

$a_8 = 7$

1	32	12	7
C	2	1	1

$a_9 = 6$

1	32	12	7
C	2	1	1

↓ ↓ ↓

1 0 0

$a_{10} = 12$

1	32	12
C	1	1

2. m = the number of elements in the stream

$m' = 3$

$k = 3$

$\frac{11 - 3}{3 + 1} = 2$

$\frac{m - m'}{k + 1}$

3 CountMin Sketch Algorithm

$$S = \{1, 1, 1, 2, 4, 4, 3, 2, 3, 2, 3\}$$

Given the following three hash functions, perform the CountMin Sketch algorithm on the data stream S above.

$$h_1(x) = (x + 1) \bmod 3$$

$$h_2(x) = (3x + 1) \bmod 3$$

$$h_3(x) = (5x + 1) \bmod 3$$

1. Present the Hash table.
2. Present the Counter Matrix.
3. Present the estimated frequency of each element.
4. Present the difference to the true frequency.

①

$S=1 \quad (2, 1, 0) \quad \times 3 \quad 6$

$S=2 \quad (0, 1, 2) \quad \times 3$

$S=3 \quad (1, 1, 1) \quad \times 3$

$S=4 \quad (2, 1, 0) \quad \times 2$

②

	0	1	2	3	4
h_1	3	3	5	0	0
h_2	0	11	0	0	0
h_3	5	3	3	0	0

③

$S=1 \quad 5, 11, 5 \rightarrow \min, 5$

$S=2 \quad 3, 11, 3 \rightarrow 3$

$S=3 \quad 3, 11, 3 \rightarrow 3$

$S=4 \quad 5, 11, 5 \rightarrow 5$

④

true,

3 3 3 2.

estimate

5 3 3 5.

↑ - - - ↑

over estimate

4 Count Sketch Algorithm

$$S = \{1, 1, 1, 2, 4, 4, 3, 2, 3, 2, 3\}$$

$$h_1(x) = (x + 1) \bmod 3$$

$$h_2(x) = (3x + 1) \bmod 3$$

$$h_3(x) = (5x + 1) \bmod 3$$

Using the hash functions above, perform the Count Sketch algorithm with the following sign hash functions:

$$s_1(x) = ((2x + 1) \bmod 3) \bmod 2$$

$$s_2(x) = ((3x + 2) \bmod 3) \bmod 2$$

$$s_3(x) = ((5x + 2) \bmod 3) \bmod 2$$

$$s_i(x) = \begin{cases} 0, & +1 \\ 1, & -1 \end{cases}$$

1. Present the Hash table.
2. Present the Counter Matrix.
3. Present the estimated frequency of each element.
4. Present the difference to the true frequency.

①

	h_1	h_2	h_3	
1	(2, 1, 0)	$\times 3$	(2)	
2	(0, 1, 2)	$\times 3$		
3	(1, 1, 1)	$\times 3$		
4	(2, 1, 0)	$\times 2$		

②

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	0	0	1
2	0	0	0
3	1	0	0
4	0	0	1

③

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

④

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑤

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑥

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑦

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑧

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑨

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑩

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑪

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑫

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑬

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑭

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑮

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑯

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑰

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑱

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑲

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

⑳

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉑

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉒

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉓

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉔

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉕

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉖

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉗

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉘

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉙

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉚

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉛

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉜

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉝

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉞

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉟

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊱

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊲

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊳

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊴

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊵

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊶

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊷

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊸

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊹

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊺

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊻

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊼

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊽

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊾

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊿

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㉟

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3	-1	1	1
4	1	1	-1

㊱

	$s_1(x)$	$s_2(x)$	$s_3(x)$
1	1	1	-1
2	1	1	1
3			

(4). est ~~11, 3, 11, 11~~
5 3 3 5.

~ ~ ~